Data Scientist Technical Exercise – Upstream Migration Interventions

Context

Asylum claimants in the UK often arrive by illegal entry routes including crossing the channel in small boats (also known as irregular migration), although a significant proportion enter the UK legally by visa and then claim asylum.

The government wants to reduce the number of people arriving on small boats. A key lever is to work upstream internationally in source and transit countries to prevent, mitigate and resolve conflict, provide humanitarian assistance at scale and support low income countries to drive sustainable and inclusive growth. Other types of interventions include but are not restricted to:

- Returns agreements with source countries to return refused asylum seekers, foreign national offenders and others who have no right to be in UK
- 1:1 exchange agreements with transit countries such as the UK-France 'one in one out' agreement and the 2016 EU-Turkey deal
- Aid programmes which support refugees to remain in source and transit companies

In order to return an asylum claimant to a source country the UK typically needs to consider and refuse their asylum claim first. Exchanges with transit countries do not usually require this, but transit countries are also not obliged to accept third country nationals. There are two main types of departure from the UK, enforced (where the government detains someone and forces them to leave the UK) and voluntary (where someone leaves the UK of their own accord). Returns and exchange agreements usually focus on enforced returns. Not everyone returned from the UK has claimed asylum – returns figures also include foreign national offenders and visa overstayers.

Task

Analyse data on irregular migration flows into the UK and Europe, and on the UK's asylum and returns system, to **make a data-driven recommendation for the top countries the UK should intervene with to reduce small boat arrivals**. It is **not** essential to consider qualitative factors such as diplomatic relations or geopolitics as part of this exercise, but you may reference these in interpreting your data if you wish.

You will need to:

1. Produce a short presentation to communicate your findings. The presentation should include data visualisations.

2. Email a zip file of your code and presentation with the subject line '[Your Name] - Data Scientist technical exercise' to mmather@no10.gov.uk at the time specified in the cover email. **Rename any code files (e.g. ".py") to text files (e.g. ".txt")** - otherwise they may be caught by security checks. Please also convert your presentation to pdf format to ensure they can be opened.

3. Present your findings to the panel. You should aim for a presentation that lasts for four minutes, and there will be a hard stop at five minutes. The target audience for this presentation is the Prime Minister, who is informed about migration flows but doesn't have an analytical background. Although the panel will include data scientists, you should pitch your original presentation as if it is to the Prime Minister. Some follow-up questions may be analytical, which you should feel free to answer with full technical detail.

Points you may wish to consider

Below are examples of some of the questions you may wish to consider. **You do not need to consider them all.** They are intended only as prompts to help you think through your analysis.

- What are the top nationalities (i.e. source countries) for those using illegal entry routes?
- Can you infer anything about the importance of different routes into Europe and their associated transit countries for small boat arrivals to the UK? You may wish to use data at nationality level.
- What are current trends, and how might the most important source and transit countries change over time?
- How does the number of returns relate to the number of people using illegal entry routes?
- Which countries is the UK struggling to return people to?

Relevant data

Data on the UK asylum system is published as part of quarterly immigration statistics: Immigration system statistics, year ending December 2025 - GOV.UK. Data tables for this publication can be found here Immigration system statistics data tables - GOV.UK. The illegal entry routes, asylum claims and initial decisions, and returns summary and detailed data tables are likely to be the most relevant to this exercise.

Frontex publish data on migration flows by nationality and European route here: Migratory Map (data download link at the bottom of the webpage).

Feel free to include other relevant data sources in your analysis.

Expectation

You are required to demonstrate:

- a general understanding of how to manipulate data
- awareness of analytical techniques and best practices
- ability to communicate insight with data visualisation
- intellectual curiosity and story-telling ability
- you are welcome to use AI, but it is not expected and you will be required to tell us how you have used AI if you do.

We do not expect an exhaustive analysis of all data. Rather, we would like to gain insight into your technical abilities and how you approach an exploratory problem from a data science / analytical perspective.

You can highlight additional or different approaches you would consider if you had more time.

Tips on prioritisation during this exercise

1. This exercise is intended to simulate a type of short-term ask that 10DS sometimes has to complete for senior decision makers. We are testing how effective you would be in this kind of situation in real-life.
    a. We weight our assessment heavily on the quality of insight provided (including the relevance of insight to decision makers and analytical robustness – it needs to be correct if being seen

by a senior decision maker), on the clarity and professionalism of the presentation, and on the quality of data visualisation.

b. We weight less heavily on things that are low impact during this kind of ask e.g. having really well-documented code – the objective is to get to the right insights as quickly as possible.

2. Ensure you leave sufficient time to develop a high-quality presentation once you have completed your analysis of the data. We recommend spending at least an hour on the presentation itself. A common mistake people make when doing this exercise is to spend all the time on analysis – in a real-life situation, your analysis will not have impact with a senior decision-maker if it is not communicated effectively.